



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES PAPER 1	DATE	24 NOVEMBER 2023
GRADE	11	MARKS	200
EXAMINER NAME	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
CLASS		DURATION	3 HOURS

MARKING GUIDELINES

Question 1:

- 1.1. F ✓
 A ✓
 C ✓
 J ✓
 K ✓
 H ✓
 E ✓
 I ✓
 D ✓
 B ✓

[10]

1.2.

1.2.1.	1.2.2.	1.2.3.	1.2.4.	1.2.5.	1.2.6.	1.2.7.	1.2.8.	1.2.9.
A ✓	B ✓	D ✓	D ✓	C ✓	C ✓	B ✓	C ✓	B ✓✓

[10]

1.3.

- 1.3.1. B ONLY ✓
 1.3.2. BOTH ✓
 1.3.3. NONE ✓
 1.3.4. NONE ✓
 1.3.5. BOTH ✓

[5]

1.4.

- 1.4.1. More oxygen ✓ is needed to be pumped around the body for increased respiration ✓ and the increase CO₂ needs to be brought to the lungs for exhalation faster. ✓ (any 2)

- 1.4.2. Learner A ✓ (1)

- 1.4.3. Their resting heart rate is lower OR their heart rate returns to resting HR quicker than B ✓ (1)

- 1.4.4. 4 minutes ✓ (1)

- 1.4.5. Heart rate (bpm) ✓ (1)

- 1.4.6. same gender (female)

Same age of learners ✓

Same apparatus to measure HR ✓✓

(any 2)

- 1.4.7. Heading (H)- all variables ✓

X-axis (X)- title + unit ✓ and scale ✓ (time)

Y-axis (Y)- title + unit ✓ and scale ✓ (HR)

Key (K) ✓

Plotting (P) ✓

(7)

[15]

- 1.5.
- 1.5.1. B ✓ (1)
- 1.5.2. Blood flow is lowered✓ due to the narrowing of the lumen ✓ (2)
- 1.5.3. Cholesterol ✓ (1)
- 1.5.4. Lowered the intake of animal fat in one's diet
Increase HDLs in one's diet ✓ (any 1)
- 1.5.5. $250 - 155 = 95$ ✓ $\text{cm}^3/\text{minute}$ ✓ (2)
- 1.5.6. 250×60 ✓ = 15 000 $\text{cm}^3/\text{minute}$ ✓ (no unit no mark) (2)
- 1.5.7.
- a. Only a small cut/ incision is made in the skin
No other vessels are used during this procedure
Only 2-3 hours is spent in hospital after the procedure as opposed to 7 days
Recovery time is 7 days as opposed to 12 weeks.
The risk of a heart attack during the procedure is lower than CABG ✓✓ (any 2)
- b. The chance of failure within a year is substantially lower
The patient is under general anesthetic (trauma and stress are lowered)
Can be used when multiple blockages are present whereas a stent can only be used for a single blockage ✓✓ (any 2)
- [13]
- 1.6.
- 1.6.1. A group of organisms of the same species ✓ that occupy the same are and can breed ✓ (2)
- 1.6.2. Time/weeks ✓ OR species A and B ✓ (1)
- 1.6.3. $100 - 20$ ✓ = 80 more of Species A ✓ (2)
- 1.6.4. Competitive exclusion ✓ (1)
- 1.6.5. Separated: numbers increase greatly due to their being no competition ✓
Together: both species grow in the first week ✓ as there is sufficient food but thereafter species A increases greatly while species B drastically decreases ✓ due to being outcompeted by A when food is limited ✓ (4)
- 1.6.6.
- a. Predator: organism that hunts and kills its food ✓
Prey: organism eaten by another (predator) ✓ (2)

b. Heading (H) ✓

Axes headings (time on X-axis, species number on Y-axis) (A) ✓

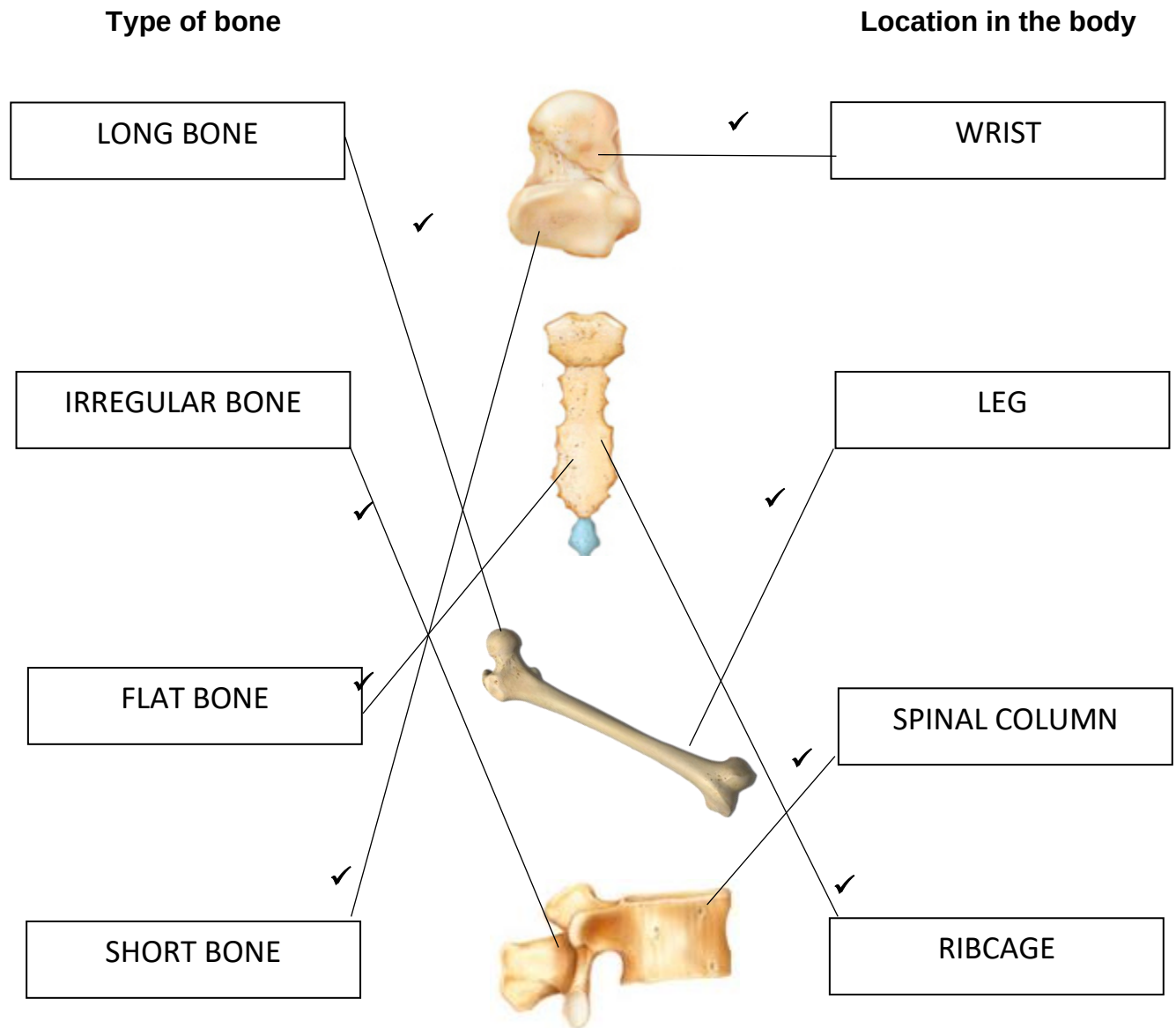
Prey numbers always higher than predator (N) ✓

Cyclic change in predator always slightly after prey (C) ✓

Key (K) ✓

(5)
[17]

1.7.



[8]

1.8. Hydrostatic ✓

Exoskeleton ✓

[2]

Total Question 1: 80 marks

Question 2

2.1.

2.1.1. Movement in correlation with muscles

Protection of internal organs

Storage of minerals

Support and structure

Production of blood cells ✓✓

(any 2)

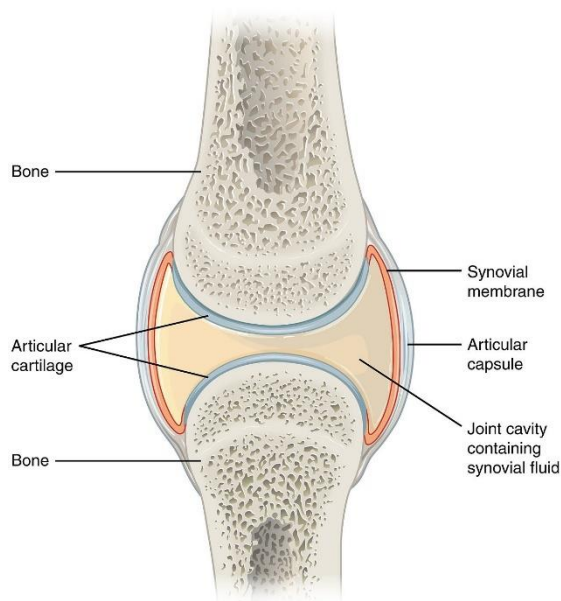
2.1.2. Pectoral ✓

(1)

2.1.3. A ✓ and B ✓

(2)

2.1.4.



Heading ✓

Correct diagram/ accuracy ✓

Any 3 labels ✓✓✓

(5)

2.1.5. H ✓

(1)

2.1.6. Ball and socket joint ✓ - allows for rotation and movement in many directions ✓

(2)

2.1.7.

B- scapula ✓

D- radius ✓

E- ulna ✓

G- metacarpals ✓

(4)

2.1.8. Produces red blood cells ✓ and white blood cells ✓

(2)

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2.2.

2.2.1. Inflammation of the joints ✓ (1)

2.2.2. A region where two or more ✓ bones meet ✓ (2)

2.2.3. Ligament ✓ (1)

2.2.4. Contains collagen ✓ and elastic fibres ✓

Strong and resilient ✓

Flexible ✓ (any 2)

2.2.5. Articular cartilage ✓ of joint wears away ✓

Bones of joint grind against each other ✓

Small outgrowths of bone are produced ✓ (any 3)

2.2.6. More years of wear and tear ✓ on the cartilage due to physical activity or past injuries ✓ (2)

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2.3.

2.3.1. Rickets ✓ (1)

2.3.2. Vitamin D ✓ (1)

2.3.3. Accept between 24-27 years ✓✓ (2)

2.3.4. A graph showing the relationship between bone mass and age ✓ of both men and women ✓ (2)

2.3.5. Men have higher bone mass than women

Women's bone mass/ density decreases more than males after the age of 40.

Men and women experience a peak in bone mass at the same age ✓✓✓✓ (any 2 x 2=4)

[10]

Total Question 2: 40 marks

Question 3

- 3.1.
- 3.1.1. A- right atrium✓
C- Left ventricle ✓ (2)
- 3.1.2.
a. Closed ✓
b. Open ✓ (2)
- 3.1.3.
a. Blood pressure is high and elastic tissue can recoil✓ to even out the pressure ✓ (2)
b. Allows one way flow of blood/ prevents the backflow of blood ✓ (1)
- 3.1.4. pulmonary artery and a pulmonary vein labelled on diagram ✓✓ (2)
- 3.1.5. Heading ✓
Any 2 differences x 2:
- Arteries have a thicker muscular layer with more elastic fibres; veins have less elastic fibres and a thinner muscular layer ✓✓
- Arteries have a narrower, regular shaped lumen; veins have a large, irregular shaped lumen✓✓
- Arteries don't contain valves; veins contain semilunar valves ✓✓
-1 Table not drawn/ drawn incorrectly (5)
- 3.1.6. SA node ✓ labelled in correct place ✓ (2)
- 3.1.7.
a. It is the aorta✓ which is the artery in which BP is measured accurately (1)
b. Pressure increase when heart is contracted (systole) ✓and decreases when its relaxed (diastole) ✓ (2)
c. Increase activity
Increased stress ✓ (any 1)
d. 105 (accept 103-107) ✓/ 68 (accept 66-70) ✓ (2)

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- 3.2.
- 3.2.1. X- 110 or 109 ✓
Y- 144 ✓ (2)
- 3.2.2. Age ✓ (1)
- 3.2.3. The lowest one's heart rate can be to break down fat stores OR the rate at which one's heart should beat to burn fat ✓✓ (2)
- 3.2.4. Same gender of cyclists
Same number of cyclists in each age group ✓ (any 1)
- 3.2.5. As age increases ✓ they minimum heart rate for metabolising fat and improving fitness decreases ✓ (2)
- [8]
- 3.3.
- 3.3.1. They are flexible allowing them to fit through narrow capillaries to deliver oxygen OR
The biconcave shape increases surface area for oxygen transportation OR
The lose their nucleus upon maturity to allow for more space to carry oxygen ✓✓ (any 1 x 2)
- 3.3.2. $250\,000\,000 \times 12\% \checkmark = 30\,000\,000 + 250\,000\,000 \checkmark = 280\,000\,000$ cells ✓ (3)
- 3.3.3. X anywhere on the major vessels ✓ (1)
- 3.3.4. High BP/ smoking/ high blood cholesterol/ obesity/ type 2 diabetes/ environmental stress/
sedentary (inactive) lifestyle/ diet high in sugars and saturated fats/ diet low in fruit and
vegetables ✓✓ (any 2)
- 3.3.5. Age/ genetic predisposition/ being male/ having a high achieving personality ✓ (any 1)
- 3.3.6. Paralysis of the limbs on one side of the body/ difficulties with speech and/or swallowing/
unconsciousness/ visual field disturbances ✓ (any 1)

Total Question 3: 40 Marks

Question 4

- 4.1.
- 4.1.1. Census ✓ (1)
- 4.1.2. A ✓ (1)
- 4.1.3. The base of the age pyramid is wide but it narrows towards the top ✓
This indicates a high proportion of individuals in the younger age group. ✓ When they grow up and reproduce the population will increase in size. ✓ The narrowing of the pyramid towards the top also indicates a high death rate with increasing age. ✓ (4)
- 4.1.4. The population size at each age group remains almost equal. ✓ This indicates that the birth and death rates are almost the same. ✓ Therefore, the population will remain more or less the same/ stable population. ✓ (3)
- 4.1.5.
- a. Exponential / geometric / logarithmic / J-shaped ✓ (1)
- b. The human population has almost reached the carrying capacity of planet earth and therefore begin to experience environmental resistance such as:
- Competition for food, water and space
 - This could lead to war or increased crime rate
 - New diseases
 - Greater awareness and education (2)
- c.
- i. The maximum number✓ of individuals an area can sustain ✓ (2)
- ii. Improving methods of food production (monoculture) ✓
Methods of treating diseases have greatly improved ✓ (2)
- iii. Could lead to possible extinction ✓ (1)
- iv. Educate and organize awareness programmes to highlight the consequences of unchecked population growth
Tax incentives for smaller families
Introduction of family planning programmes
Any other logical answer ✓ (any 1)

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- 4.2.
- 4.2.1. Predator- prey ✓ (1)
- 4.2.2. Green flies ✓ (1)
- 4.2.3. The rising and falling of the green fly population occurs before that of the lady bugs ✓ OR the number of lady birds is continually lower than that of the green flies (1)
- 4.2.4. The two species control each other's population size ✓
- When the lady bird population increased the green flies decreases ✓
 - When the green flies population decreases the lady birds decreases ✓
 - Therefore both populations do not exceed carrying capacity ✓ (3)
- 4.2.5. Due to an increase in lady bug ✓ numbers there was an increase in predation, ✓ leading to a decrease in green flies ✓ as they are the food source. (3)
- 4.3. [9]
- 4.3.1. $P = (M \times C)/R$
 $60 = (M \times 15)/5$ ✓
 $M = (60 \times 5)/15$ ✓
 = 20 snails caught in the first sample ✓ (3)
- 4.3.2. Only a short time should pass between the first and second sampling so no births and deaths occur.
- The markings must not damage the individual
- The markings must not affect the animals behaviour or movement.
- The marked animals must be given enough time to mix freely with the rest of the population before the second sample is taken.
- The population needs to be closed ✓✓ (any 2)
- 4.3.3. Repeat the sampling several times and calculate an average ✓ (1)
- 4.3.4. Lady birds are very mobile ✓ and not easily visible ✓ (2)
- 4.4.
- 4.4.1. Beak shape and size ✓ (1)
- 4.4.2. a. C ✓ b. A ✓ (2)
- 4.4.3. Yes. ✓
- They occupy different ecological niches
- The large tree finch eats mainly animal material, while the large ground finch eats mainly plant material. (any 1 ✓) (2)

Total Question 4: 40 marks